

Q1: Color Scales for Multi-Variable Data [R]

Task: Design a visualization that effectively represents Temperature, AQI, Rainfall and Humidity of 20 Indian cities using appropriate colour scales. Select a suitable colour palette and justify its effectiveness for accurate interpretation and accessibility.

R program

```
library(ggplot2)
library(patchwork)

# Data: 20 Indian cities
cities <- data.frame(
  City = c("Delhi", "Mumbai", "Chennai", "Kolkata", "Hyderabad", "Bangalore", "Pune", "Ahmedabad", "Jaipur", "Lucknow",
    "Bhopal", "Patna", "Goa", "Kochi", "Nagpur", "Indore", "Surat", "Visakhapatnam", "Coimbatore", "Mysuru"),
  Temp = c(44, 35, 38, 40, 39, 31, 33, 43, 45, 39, 38, 40, 32, 30, 41, 37, 36, 34, 29, 28),
  Humidity = c(22, 84, 78, 72, 48, 66, 58, 26, 18, 42, 36, 45, 89, 91, 30, 34, 76, 82, 68, 64),
  Rainfall = c(6, 280, 220, 190, 85, 120, 95, 2, 1, 35, 48, 72, 360, 410, 12, 20, 180, 240, 130, 145),
  AQI = c(365, 110, 125, 180, 160, 72, 84, 290, 320, 240, 175, 220, 58, 52, 270, 190, 145, 98, 60, 55)
)

# Order cities by AQI so the pattern between the variables is easy to see
cities$City <- factor(cities$City, levels = cities$City[order(cities$AQI)])

# One heatmap column per variable, each with its own sequential colour scale
panel <- function(var, label, palette, show_city = FALSE) {
  d <- data.frame(City = cities$City, value = cities[[var]], x = label)
  r <- range(d$value)
  lim <- c(r[1] - 0.2 * (r[2] - r[1]), r[2]) # start the scale slightly above the palette's near-white end
  d$txt <- ifelse((d$value - lim[1]) / (lim[2] - lim[1]) > 0.62, "white", "black")
  p <- ggplot(d, aes(x = x, y = City, fill = value)) +
    geom_tile(colour = "white", linewidth = 0.6) +
    geom_text(aes(label = value, colour = txt), size = 2.8) +
    scale_colour_identity() +
    scale_fill_distiller(palette = palette, direction = 1, name = NULL, limits = lim,
      guide = guide_colourbar(barwidth = unit(2.4, "cm"), barheight = unit(0.25, "cm"))) +
    labs(x = NULL, y = NULL) +
    theme_minimal(base_size = 9) +
    theme(panel.grid = element_blank(), legend.position = "bottom",
      axis.text.x = element_text(face = "bold"))
  if (!show_city) p <- p + theme(axis.text.y = element_blank())
  p
}

p <- panel("Temp", "Temperature (deg C)", "YlOrRd", TRUE) +
  panel("Humidity", "Humidity (%)", "Blues") +
  panel("Rainfall", "Rainfall (mm)", "Greens") +
  panel("AQI", "AQI", "Purples") +
  plot_layout(nrow = 1) +
  plot_annotation(title = "Climate and Air Quality of 20 Indian Cities (sorted by AQI, lowest at bottom)")
print(p)

# Relationships between the four variables
round(cor(cities[, c("Temp", "Humidity", "Rainfall", "AQI")]), 2)
```

Output (correlation between the four variables)

	Temp	Humidity	Rainfall	AQI
Temp	1.00	-0.73	-0.62	0.94
Humidity	-0.73	1.00	0.93	-0.85
Rainfall	-0.62	0.93	1.00	-0.73
AQI	0.94	-0.85	-0.73	1.00

Visualization

Climate and Air Quality of 20 Indian Cities (sorted by AQI, lowest at bottom)



Chart design

A heatmap table: one row per city and one column per variable. Cities are sorted by AQI, so the other columns can be compared with AQI row by row. The exact value is printed inside every cell, so colour shows the pattern and the number gives the accurate reading.

Colour scales and palette

Each variable has its own **sequential single-hue scale**: Temperature uses yellow to red, Humidity uses light to dark blue, Rainfall uses light to dark green, and AQI uses light to dark purple. All four variables are ordered numbers with no meaningful middle value, so a sequential scale (not diverging or categorical) is the right type. Using a different hue per variable stops the reader from confusing one column with another, and each column has its own legend with its real units.

Accessibility

Every scale changes mainly in **lightness** (dark means high), so the chart is still readable in greyscale and for colour-blind viewers; red and green are never used together as a contrast pair. The scales start slightly above the palette's near-white end so that the lowest values are still visible against the white background. The cell text switches between black and white to keep a strong contrast with the fill, and the numbers repeat the information in colour.

What the chart shows

The Temperature column darkens in step with AQI ($r = 0.94$): the hot cities Delhi (365), Jaipur (320) and Ahmedabad (290) have the worst air quality. Humidity and Rainfall are high together ($r = 0.93$) in coastal cities such as Kochi, Goa and Mumbai, which have the cleanest air (AQI 52, 58 and 110), giving negative correlations with AQI (-0.85 and -0.73). These are associations only, not proof of cause.

Q2: Cartesian vs. Polar Coordinates [R]

Task: Using the monthly electricity generation data, develop visualizations using Cartesian and Polar coordinates. Compare both and determine which coordinate system provides better analytical insight. Explain the design decisions.

R program

```
library(ggplot2)
library(patchwork)

# Data: monthly electricity generation by source
gen <- data.frame(
  Month = 1:12,
  Coal = c(620, 580, 590, 605, 640, 670, 690, 720, 700, 680, 650, 630),
  Gas = c(310, 295, 305, 320, 340, 355, 360, 370, 365, 350, 335, 320),
  Solar = c(45, 60, 90, 130, 180, 230, 250, 245, 210, 160, 90, 50),
  Wind = c(80, 95, 120, 150, 170, 185, 195, 190, 175, 150, 110, 90),
  Hydro = c(140, 150, 160, 170, 180, 195, 205, 210, 200, 185, 165, 150)
)

long <- reshape(gen, direction = "long", varying = c("Coal", "Gas", "Solar", "Wind", "Hydro"),
  v.names = "Generation", timevar = "Source",
  times = c("Coal", "Gas", "Solar", "Wind", "Hydro"), idvar = "Month")
long$Source <- factor(long$Source, levels = c("Coal", "Gas", "Solar", "Wind", "Hydro"))
cols <- c(Coal = "#333333", Gas = "#0072B2", Solar = "#E69F00", Wind = "#009E73", Hydro = "#CC79A7")

# (a) Cartesian coordinates: line chart
p_cart <- ggplot(long, aes(x = Month, y = Generation, colour = Source)) +
  geom_line(linewidth = 1) + geom_point(size = 1.8) +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  scale_colour_manual(values = cols) +
  labs(title = "(a) Cartesian coordinates", x = "Month", y = "Generation") +
  theme_minimal(base_size = 10)

# (b) Polar coordinates: the same lines wrapped around a 12-month circle
closed <- rbind(long, transform(long[long$Month == 1, ], Month = 13)) # repeat Jan so each line closes
p_polar <- ggplot(closed, aes(x = Month, y = Generation, colour = Source)) +
  geom_line(linewidth = 1) + geom_point(size = 1.8) +
  coord_polar(theta = "x") +
  scale_x_continuous(limits = c(1, 13), breaks = 1:12, labels = month.abb) +
  scale_colour_manual(values = cols) +
  labs(title = "(b) Polar coordinates", x = NULL, y = NULL) +
  theme_minimal(base_size = 10)

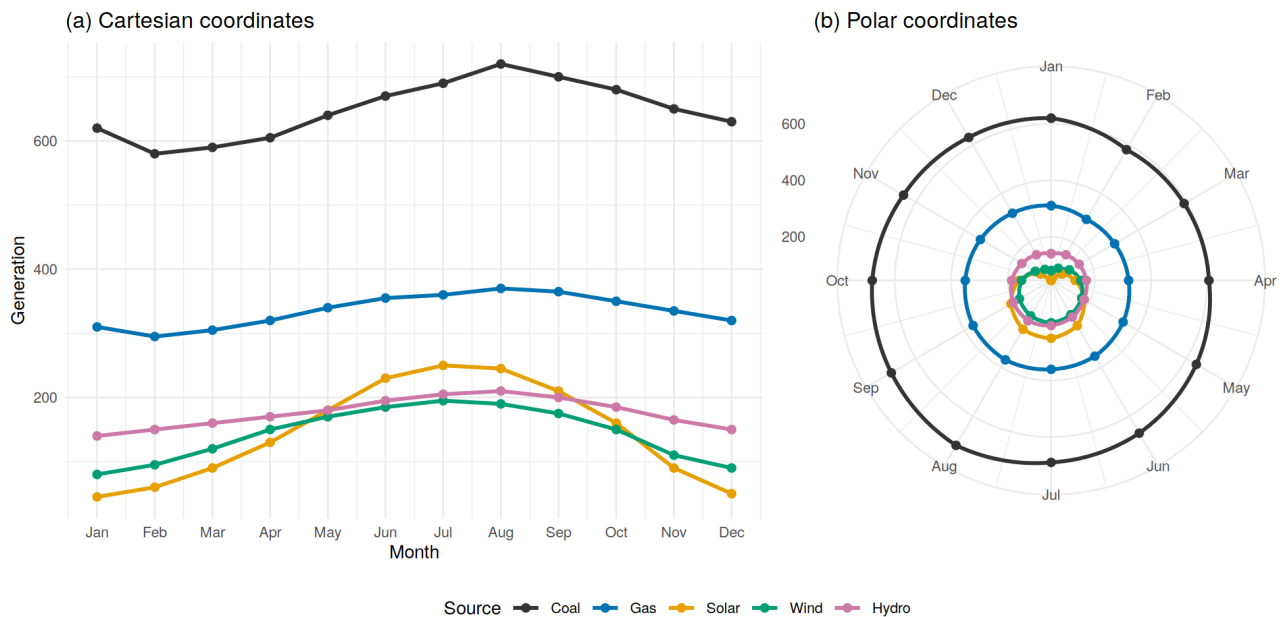
print(p_cart + p_polar + plot_layout(guides = "collect") &
  theme(legend.position = "bottom"))

# Summary of each source: mean, peak, month of peak, trough, seasonal swing (peak / trough)
do.call(rbind, lapply(split(long, long$Source), function(d)
  data.frame(Mean = round(mean(d$Generation)), Peak = max(d$Generation),
    PeakMonth = month.abb[d$Month[which.max(d$Generation)]],
    Low = min(d$Generation), Swing = round(max(d$Generation) / min(d$Generation), 2))))
```

Output (summary of each energy source)

	Mean	Peak	PeakMonth	Low	Swing
Coal	648	720	Aug	580	1.24
Gas	335	370	Aug	295	1.25
Solar	145	250	Jul	45	5.56
Wind	142	195	Jul	80	2.44
Hydro	176	210	Aug	140	1.50

Visualization



Design decisions

The data is a time series over 12 months, so a line chart with points is used in both coordinate systems. Both charts use the same colour for each source (a colour-blind-safe palette), the same month order, and one shared legend, so they can be compared directly. In the polar chart the month is the angle (January at the top) and generation is the distance from the centre; January is repeated at the end so each line closes into a loop. The scale starts at zero in both charts.

Comparison of the two coordinate systems

Aspect	Cartesian (a)	Polar (b)
Reading exact values	Easy: values are read against a straight y-axis.	Hard: the reader must judge distance from the centre along a curve.
Comparing sources	Gaps between lines are clear; Solar crossing Wind and Hydro is easy to see.	The small sources (Solar, Wind, Hydro) are squeezed near the centre and overlap.
Seasonal cycle	Shown as a rise and fall, with Dec and Jan at opposite ends.	Dec joins Jan, so the yearly cycle is natural; the summer bulge (Jun to Aug) is visible.
Fairness of shapes	Equal changes look equal.	Outer values look larger than inner ones, so large sources look more important.

Conclusion

The **Cartesian** coordinate system gives better analytical insight for this data. It shows that Coal is always the largest source (580 to 720), that Coal and Gas are stable (peak is only about 1.25 times the low), and that Solar is by far the most seasonal source (45 in January to 250 in July, a swing of 5.6 times). It also shows that Solar goes above Wind and Hydro from June to September and falls below both by November. The polar chart is useful only to emphasise that the data repeats every year; it is less accurate for reading values or comparing sources.

Q3: Nonlinear Axis for Skewed Data [R]

Task: A software company has recorded project statistics for twenty software development teams. Design a visualization that requires the use of a nonlinear axis transformation to represent highly skewed data. Justify the selected transformation and explain how it improves interpretation over a linear scale.

R program

```
library(ggplot2)
library(patchwork)

# Data: 20 software teams
teams <- data.frame(
  Team = paste0("T", 1:20),
  LinesOfCode = c(12000,18000,25000,35000,48000,62000,81000,96000,120000,150000,
    185000,240000,310000,390000,480000,620000,810000,1050000,1350000,1800000),
  Bugs = c(35,52,60,72,88,110,125,142,165,188,205,240,280,320,365,420,490,560,640,720),
  Devs = c(5,7,9,10,12,14,16,18,20,22,25,28,30,34,38,42,46,50,55,60),
  Complexity = c(2.1,2.4,2.7,3,3.2,3.5,3.8,4.1,4.5,4.8,5.1,5.4,5.8,6.2,6.6,7.1,7.6,8,8.5,9)
)

base <- ggplot(teams, aes(x = LinesOfCode, y = Bugs, colour = Complexity)) +
  geom_point(size = 3) +
  scale_colour_viridis_c(option = "D", end = 0.9) +
  theme_minimal(base_size = 10)

# (a) Linear axes
p_lin <- base +
  scale_x_continuous(labels = scales::comma) +
  labs(title = "(a) Linear axes", x = "Lines of Code", y = "Bugs")

# (b) Logarithmic (base 10) axes
p_log <- base +
  geom_smooth(aes(group = 1), method = "lm", se = FALSE, colour = "grey40", linewidth = 0.7) +
  scale_x_log10(labels = scales::comma, breaks = c(1e4, 3e4, 1e5, 3e5, 1e6)) +
  scale_y_log10(breaks = c(30, 50, 100, 200, 400, 700)) +
  annotation_logticks(sides = "bl", size = 0.3) +
  labs(title = "(b) Log10 axes", x = "Lines of Code (log scale)", y = "Bugs (log scale)")

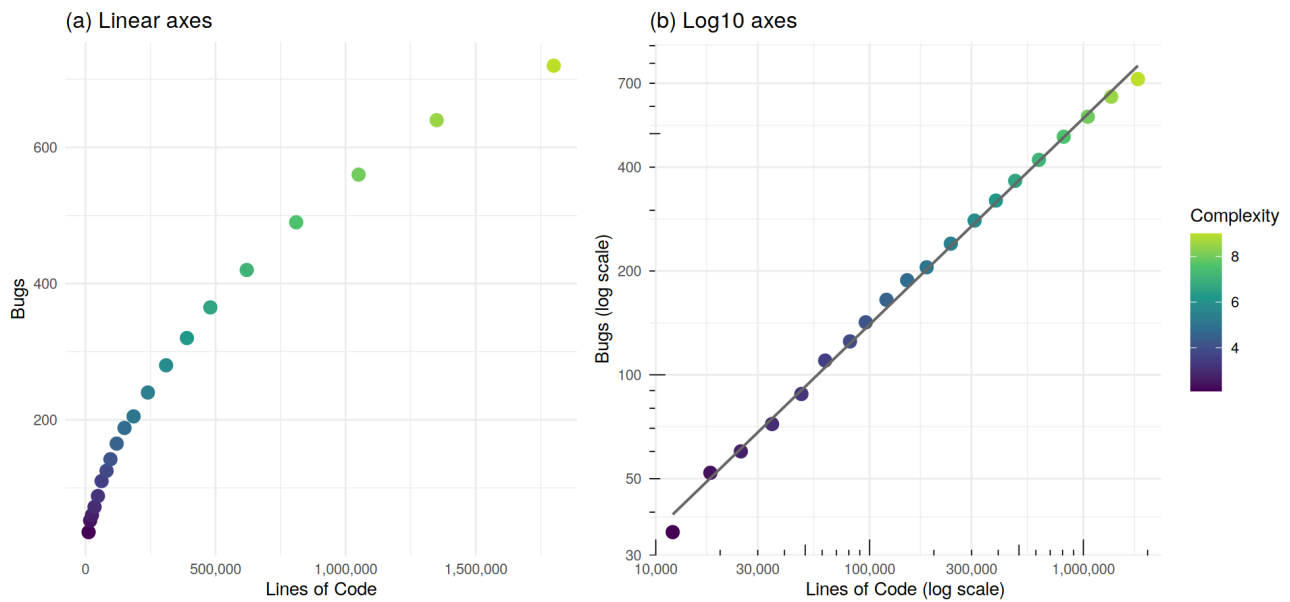
print(p_lin + p_log + plot_layout(guides = "collect"))

# Skewness and fit on the two scales
cat("Range of LinesOfCode (max/min):", max(teams$LinesOfCode) / min(teams$LinesOfCode), "\n")
cat("Teams with LinesOfCode < 100,000:", sum(teams$LinesOfCode < 1e5), "of 20\n")
skew <- function(x) mean((x - mean(x))^3) / sd(x)^3 * (1 - 1/length(x))^1.5 # moment skewness
cat("Skewness of LinesOfCode, linear :", round(skew(teams$LinesOfCode), 2), "\n")
cat("Skewness of LinesOfCode, log10 :", round(skew(log10(teams$LinesOfCode)), 2), "\n")
cat("R-squared, linear fit :", round(summary(lm(Bugs ~ LinesOfCode, teams))$r.squared, 3), "\n")
fit <- lm(log10(Bugs) ~ log10(LinesOfCode), teams)
cat("R-squared, log-log fit:", round(summary(fit)$r.squared, 3), "\n")
cat("Log-log slope:", round(coef(fit)[2], 2), "\n")
```

Output (skewness and fit on linear and log scales)

```
Range of LinesOfCode (max/min): 150
Teams with LinesOfCode < 100,000: 8 of 20
Skewness of LinesOfCode, linear : 1.35
Skewness of LinesOfCode, log10 : -0.08
R-squared, linear fit : 0.938
R-squared, log-log fit: 0.997
Log-log slope: 0.6
```

Visualization



Design decisions

A scatter plot of Lines of Code (x) against Bugs (y) is used, with the point colour showing Complexity on a sequential viridis scale. Both axes use a **log10 transformation** (scale_x_log10 and scale_y_log10) with tick labels written in the original units (10,000, 100,000, 1,000,000) and log tick marks, so the reader does not have to convert log values. A linear panel is placed next to it for comparison.

Why a logarithmic transformation

Lines of Code ranges from 12,000 to 1,800,000 (150 times) and is strongly right-skewed (skewness 1.35): 8 of the 20 teams have fewer than 100,000 lines. On the linear axis these teams are packed into the lower-left corner and cannot be told apart. Software size also changes by multiplication (double, triple), not by fixed amounts, so equal ratios should be equal distances. After the log10 transform the skewness is -0.08, so the points are spread evenly across the axis. Log10 is used because its powers of ten (10,000; 100,000; 1,000,000) are easy to read.

How it improves interpretation

On the log-log chart the points fall on a straight line (R-squared 0.997, compared with 0.938 for a straight line on the linear axes, which actually curves). The slope is about 0.60, which means that doubling the code size increases bugs by about 52% (2 to the power 0.6), so bugs grow more slowly than code size. This pattern, and the small teams, cannot be seen on the linear chart. All values are positive, so the log scale is valid. Readers should note that log axes compress large differences, so the axis is labelled 'log scale' on both panels.

Q4: Multi-Aesthetic Design [R]

Task: A multinational retail company has collected annual sales information from its branches across different regions. Design an effective visualization that simultaneously represents Sales, Number of Customers, Product Category and Region. Select suitable aesthetic mappings, apply an appropriate scale transformation wherever necessary and justify every visualization design choice.

R program

```
library(ggplot2)

# Data: 20 retail branches
shops <- data.frame(
  Branch = paste0("B", 1:20),
  Region = rep(c("North", "South", "East", "West"), 5),
  Category = rep(c("Electronics", "Furniture", "Clothing", "Food"), 5),
  Sales = c(450000, 620000, 180000, 920000, 350000, 720000, 250000, 1080000, 560000, 670000,
            195000, 980000, 430000, 760000, 280000, 1150000, 510000, 690000, 220000, 1250000),
  Profit = c(45000, 95000, -12000, 185000, 25000, 120000, 18000, 220000, 72000, 98000,
            -8000, 205000, 51000, 132000, 24000, 250000, 69000, 115000, 15000, 285000),
  Customers = c(1200, 2500, 980, 4200, 1800, 3000, 1250, 5200, 2100, 3300,
               1100, 4700, 1750, 3600, 1450, 5600, 2200, 3900, 1600, 6100)
)

cols <- c(Electronics = "#0072B2", Furniture = "#E69F00", Clothing = "#CC79A7", Food = "#009E73")

# Sales (x) vs Customers (y), colour = Category, shape = Region, both axes on a log10 scale
p <- ggplot(shops, aes(x = Sales, y = Customers, colour = Category, shape = Region)) +
  geom_point(size = 4, alpha = 0.9) +
  scale_x_log10(labels = scales::comma, breaks = c(2e5, 3e5, 5e5, 1e6)) +
  scale_y_log10(labels = scales::comma, breaks = c(1000, 2000, 4000, 6000)) +
  annotation_logticks(sides = "bl", size = 0.3) +
  scale_colour_manual(values = cols) +
  scale_shape_manual(values = c(North = 16, South = 17, East = 15, West = 18)) +
  labs(title = "Branch Sales vs Customers by Product Category and Region",
       x = "Sales (log scale)", y = "Number of Customers (log scale)") +
  theme_minimal(base_size = 10)
print(p)

# Region and Category: are they independent?
table(shops$Region, shops$Category)

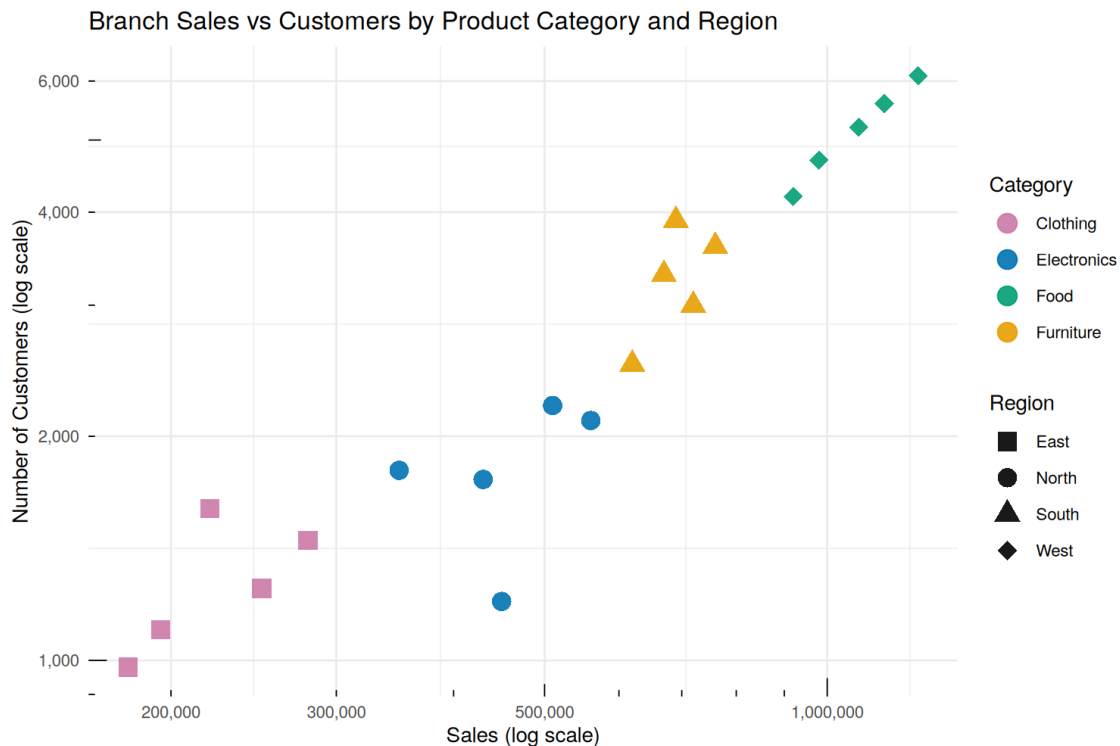
# Totals per Region/Category
agg <- aggregate(cbind(Sales, Customers) ~ Region + Category, shops, sum)
agg$SalesPerCustomer <- round(agg$Sales / agg$Customers)
agg
```

Output (Region vs Category table and totals)

		Clothing	Electronics	Food	Furniture
East		5	0	0	0
North		0	5	0	0
South		0	0	0	5
West		0	0	5	0

Region	Category	Sales	Customers	SalesPerCustomer
1 East	Clothing	1125000	6380	176
2 North	Electronics	2300000	9050	254
3 West	Food	5380000	25800	209
4 South	Furniture	3460000	16300	212

Visualization



Variable	Type	Aesthetic mapping	Why
Sales	Continuous, positive	x position, log10 scale	Position is the most accurate visual channel for numbers.
Number of Customers	Continuous, positive	y position, log10 scale	Same reason; the two numbers can be compared as a pair.
Product Category	Nominal (4 levels)	Colour hue	Hue separates groups with no order; the Okabe-Ito palette is colour-blind-safe.
Region	Nominal (4 levels)	Point shape	Shape is a second group channel that works in greyscale and for colour-blind viewers.

Scale transformation

Both axes use a log10 scale. Sales runs from 180,000 to 1,250,000 (about 7 times) and Customers from 980 to 6,100 (about 6 times), and the low-sales branches (Clothing) would be crowded together on linear axes. On a log scale equal percentage differences have equal distance, and a straight band of points shows that sales rise roughly in proportion to customers. Both variables are strictly positive, so the log is valid, and the labels show the original values. Profit is not mapped because it is not requested and has negative values (B3 and B11), which cannot be shown on a log scale.

Other design choices

A scatter plot is used because there are 20 branches and two numeric variables. Point size is kept constant so that it does not suggest an extra variable. The legends are placed on the right, outside the plot area, with axis titles that mention the log scale, and light grid lines with log tick marks help to read values.

Findings and an important limitation

Customers increase with Sales in every group, and the four groups form clear bands: Clothing (East) has the lowest sales (180,000 to 280,000), then Electronics (North, 350,000 to 560,000), Furniture (South, 620,000 to 760,000) and Food (West, 920,000 to 1,250,000, up to 6,100 customers). Sales per customer are highest for Electronics (about 254) and lowest for Clothing (about 176); Furniture and Food are about 212 and 209. However, the table above shows that every Region sells exactly one Product Category (North is Electronics, South is Furniture, East is Clothing, West is Food). So colour and shape show the same grouping, which is intentional repeated encoding for accessibility, but it also means that the effect of Region cannot be separated from the effect of Category in this data.